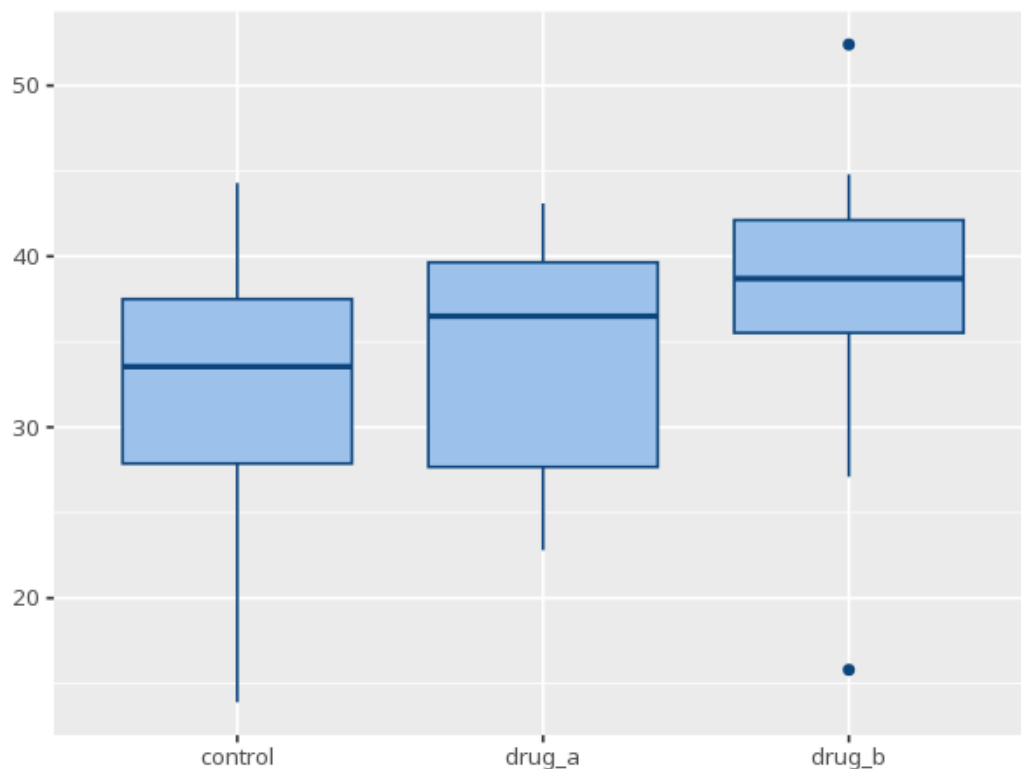


1 Kruskal-Wallis

Group	N	Median	IQR	Mean rank
control	20	33.55	9.63	24.90
drug_a	20	36.50	11.97	28.15
drug_b	20	38.70	6.60	38.45

H	df	p	ϵ^2 [95% CI]
6.56	2	.038	0.11 [0.03, 1.00]

Pair	Z	padj
control - drug_a	0.59	.556
control - drug_b	2.45	.042
drug_a - drug_b	1.87	.124



The nonparametric counterpart to one-way ANOVA. The H/p tests whether any group tends to have systematically higher or lower values (stochastic dominance) — read it as a "median difference" only when the groups have similar distribution shapes. If significant, Dunn's post-hoc shows which pairs differ, with adjusted p-values. ϵ^2 is the effect size. Your significance threshold (α) is 0.05.

A Kruskal-Wallis test gave $H(2)=6.56$, $p = .038$, $\varepsilon^2=.11$ [.03, 1.00].

Statistical basis: Kruskal-Wallis H test (nonparametric counterpart to one-way ANOVA) (Kruskal, W. H., Wallis, W. A. (1952). "Use of ranks in one-criterion variance analysis." Journal of the American Statistical Association, 47(260), 583-621.); Dunn's post-hoc pairwise comparisons (Kassambara A (2023). rstatix: Pipe-Friendly Framework for Basic Statistical Tests. R package.); ε^2 effect size (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.).